

Computer Applications in Modern Information Societies (2)

Module 1: Programming Fundamentals

Flowcharts

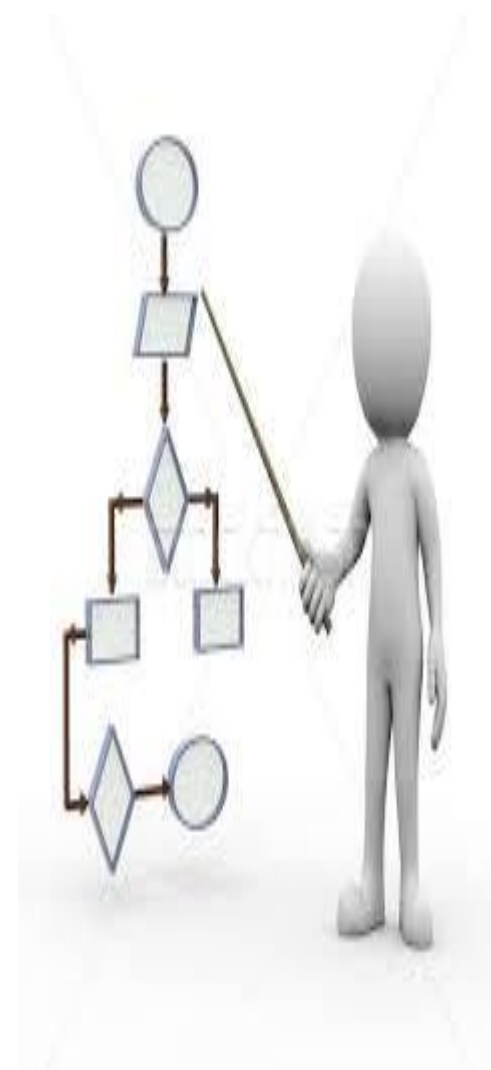
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Flowcharts

- A **flowchart**  visual or graphical representation of an algorithm.
- A flowchart is a **diagram** made up of boxes, diamonds and other **shapes**, connected by **arrows**. **Each shape** represents a step in the process, and the **arrows** show the order in which they occur.
- The flowcharts help to analyze the problem and plan its solution in a **systematic** and **orderly** manner.
- A flowchart when **translated** into a proper computer language, results in a **complete program**.

Advantages of Flowcharts

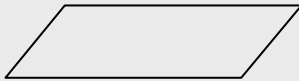
1. The flowchart shows the logic of a problem displayed in pictorial fashion which facilitates **easier checking of an algorithm**.
2. The Flowchart is good means of **communication** to other users. It is also a **compact** means of recording an algorithm solution to a problem.
3. The flowchart allows the problem solver to **break the problem into parts**. These parts can be connected to make **master chart**.
4. The flowchart is a **permanent record** of the solution which can be consulted at a later time.



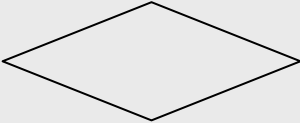
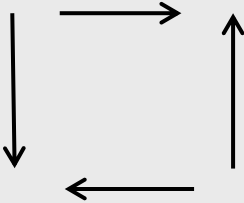
Differences between Algorithm and Flowchart

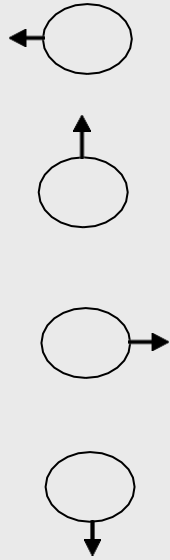
Algorithm	Flowchart
• A method of representing the step-by-step logical procedure for solving a problem.	• A diagrammatic representation of an algorithm. It is constructed using different types of boxes and symbols.
• It contains step-by-step English descriptions, each step representing a particular operation leading to solution of problem.	• The flowchart employs a series of blocks and arrows, each of which represents a particular step in an algorithm.
• These are particularly useful for small problems.	• These are useful for detailed representations of complicated programs.
• For complex programs, algorithms prove to be Inadequate.	• For complex programs, Flowcharts prove to be adequate.

Symbols used in Flowcharts

Symbol	Name	Function
	Oval	used to indicate either START/ STOP of the program.
	Input and output indicators	Parallelograms are used to represent input and output operations. Statements like INPUT, READ and PRINT are represented in these Parallelograms.
	Process Indicators	Rectangle is used to indicate any set of processing operation such as for storing arithmetic operations .

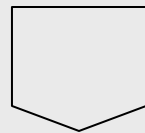
Symbols used in Flowcharts

	Decision Makers	The diamond is used for indicating the step of decision making and therefore known as decision box. Decision boxes are used <u>to test the conditions</u> or <u>ask questions</u> and depending upon the answers, the appropriate actions are taken by the computer.
	Flow Lines	Flow lines indicate the direction being followed in the flowchart. In a Flowchart, every line must have an arrow on it to indicate the direction. The arrows maybe in any direction.



On- Page connectors

Circles are used to **join the different parts** of a flowchart and these circles are called **on-page connectors**. The uses of these connectors give a **neat shape** to the flowcharts. In complicated problems, a flowchart **may run in to several pages**. The parts of the flowchart on different pages are to be joined with each other. The parts to be joined are indicated by the circle.



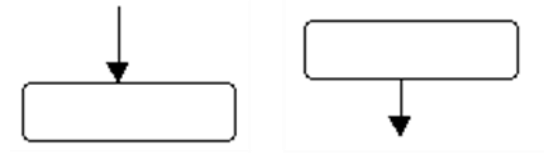
Off-page connectors

This connector represents a **break** in the path of flowchart which is too large to fit on a single page. **It is similar to on-page connector**. The connector symbol marks where the algorithm ends on the first page and where it continues on the second.

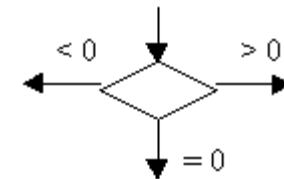
General Rules for flowcharting

1. All boxes of the flowchart are connected with **Arrows**. (Not lines)

2. Flowchart symbols have an **entry point** on the top of the symbol with no other entry points. The **exit point** for all flowchart symbols is on the bottom except for the Decision symbol.



3. The **Decision symbol** has **two exit points**; these can be on the sides or the bottom and one side.



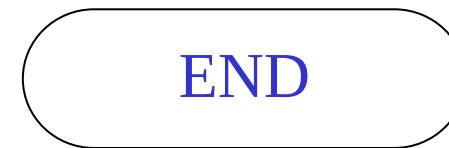
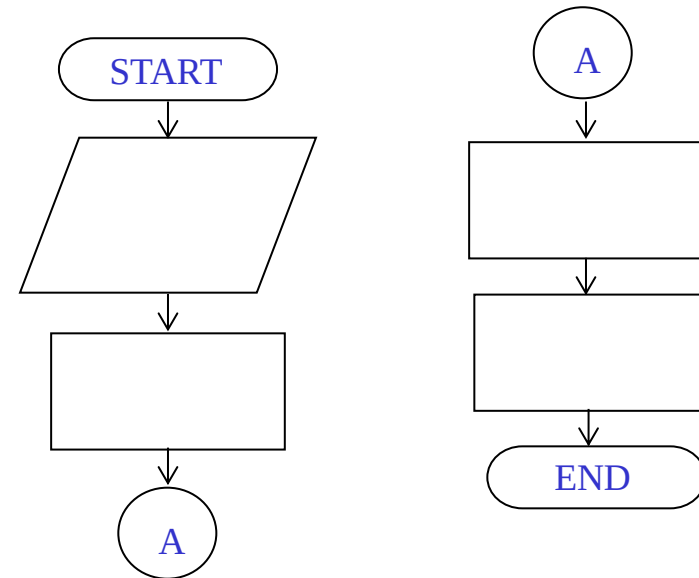
4. Generally a flowchart will flow from **top to bottom**.

General Rules for flowcharting

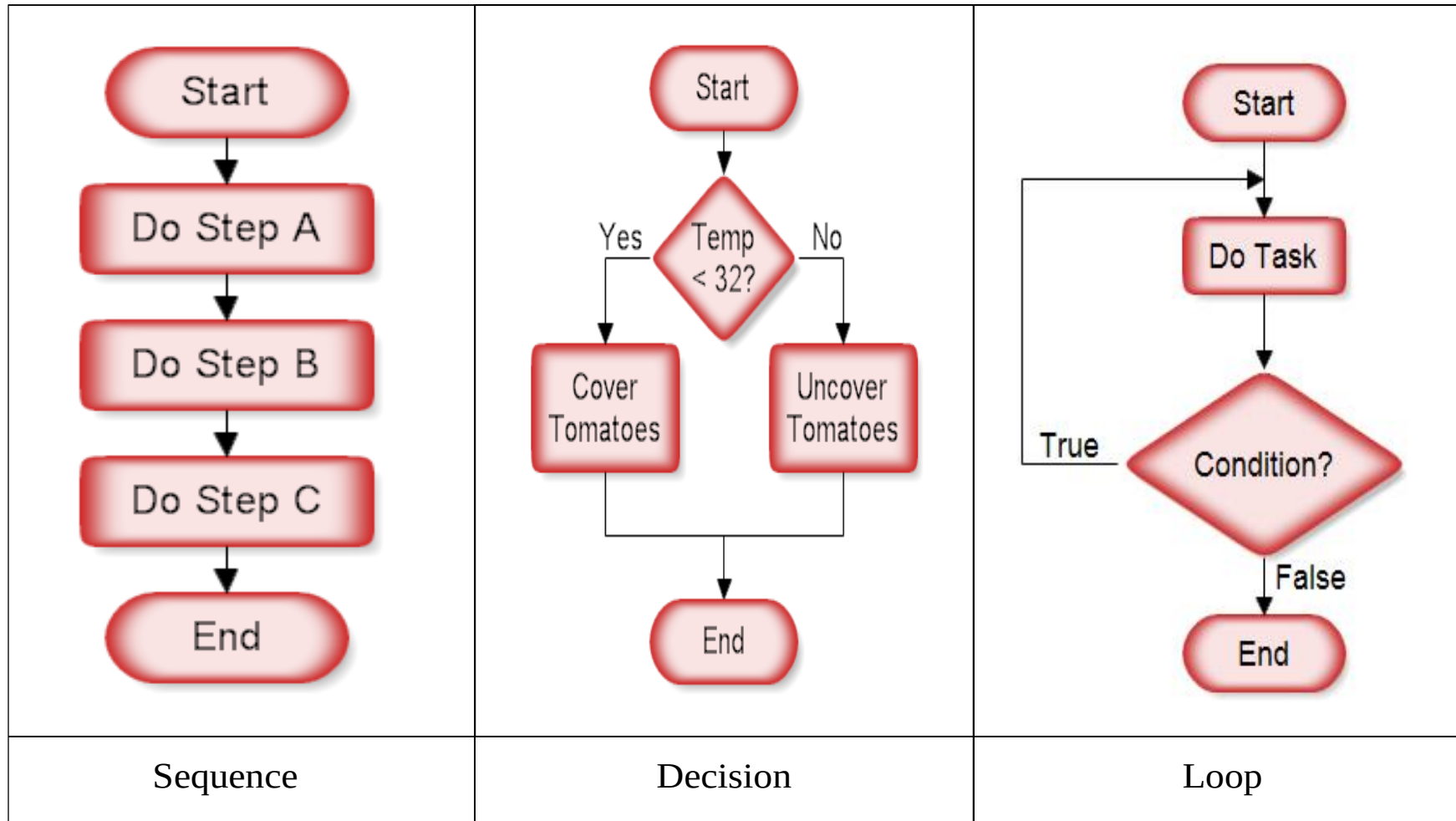
5. Connectors are used to connect breaks in the flowchart. Examples are:

- From one page to another page.
- From the bottom of the page to the top of the same page.
- An upward flow of more than 3 symbols

6. All flowcharts **end** with a terminal.



Flowcharts' main structures

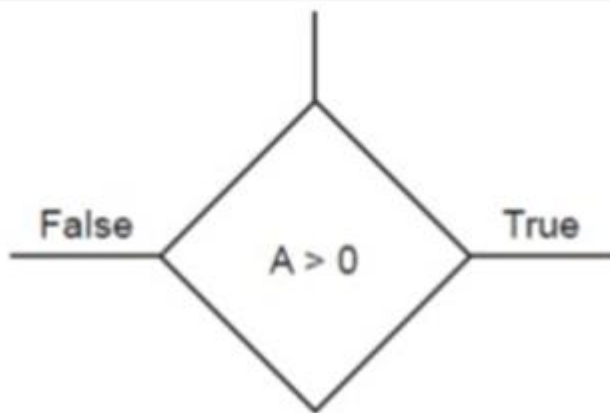


Decision Binary(structure)

- **Note:** there are **two acceptable ways to represent a decision** in all of the structures. Either method is acceptable.

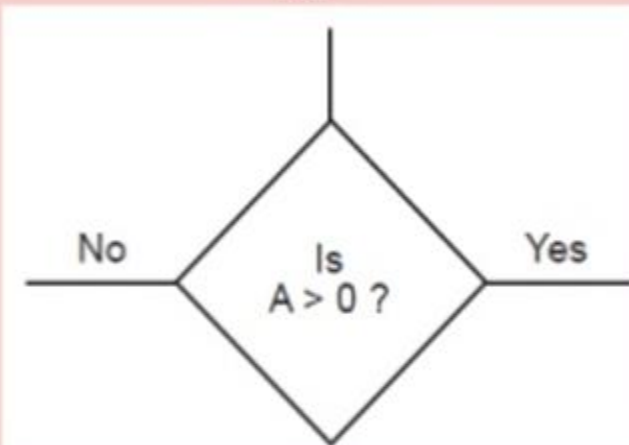
1. The **condition** is expressed as a **statement** and the two possible outcomes are indicated by

- True
- False



2. The **condition** is expressed as a **question** and the two possible outcomes are indicated by

- Yes
- No



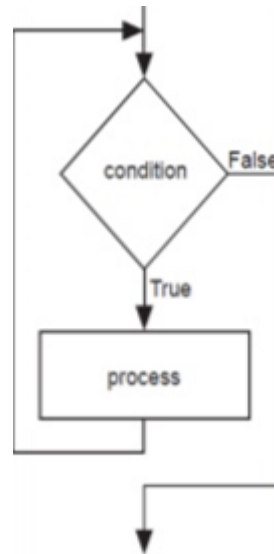
Repetition Structure

An essential feature of repetition is that each loop has a termination condition to stop the repetition.

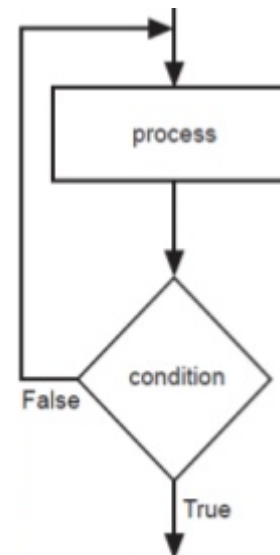
The **termination condition** can be checked or **tested**

1. At the **beginning** and is known as a **pre-test loop** or
2. At the **end** of the loop and is known as a **post-test loop**.

Pre-test Repetition
flowchart

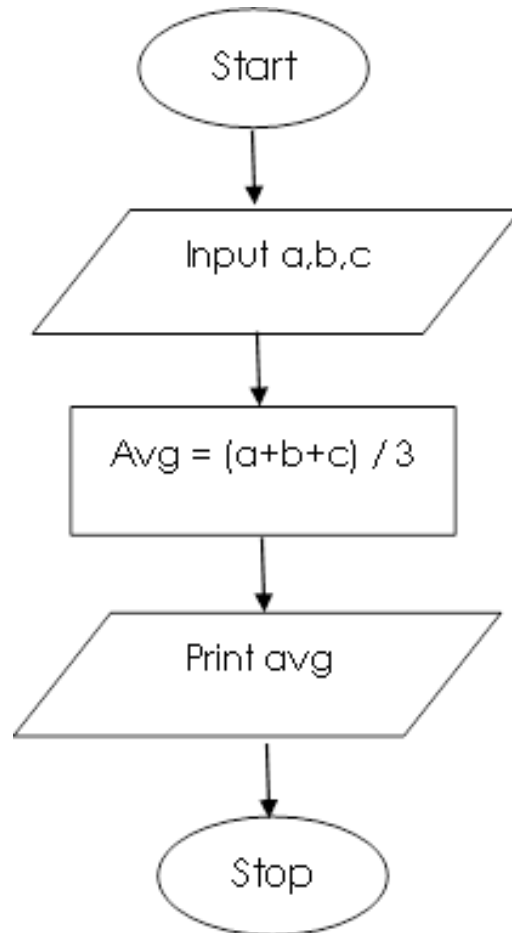


Post-test Repetition
flowchart



Example 1

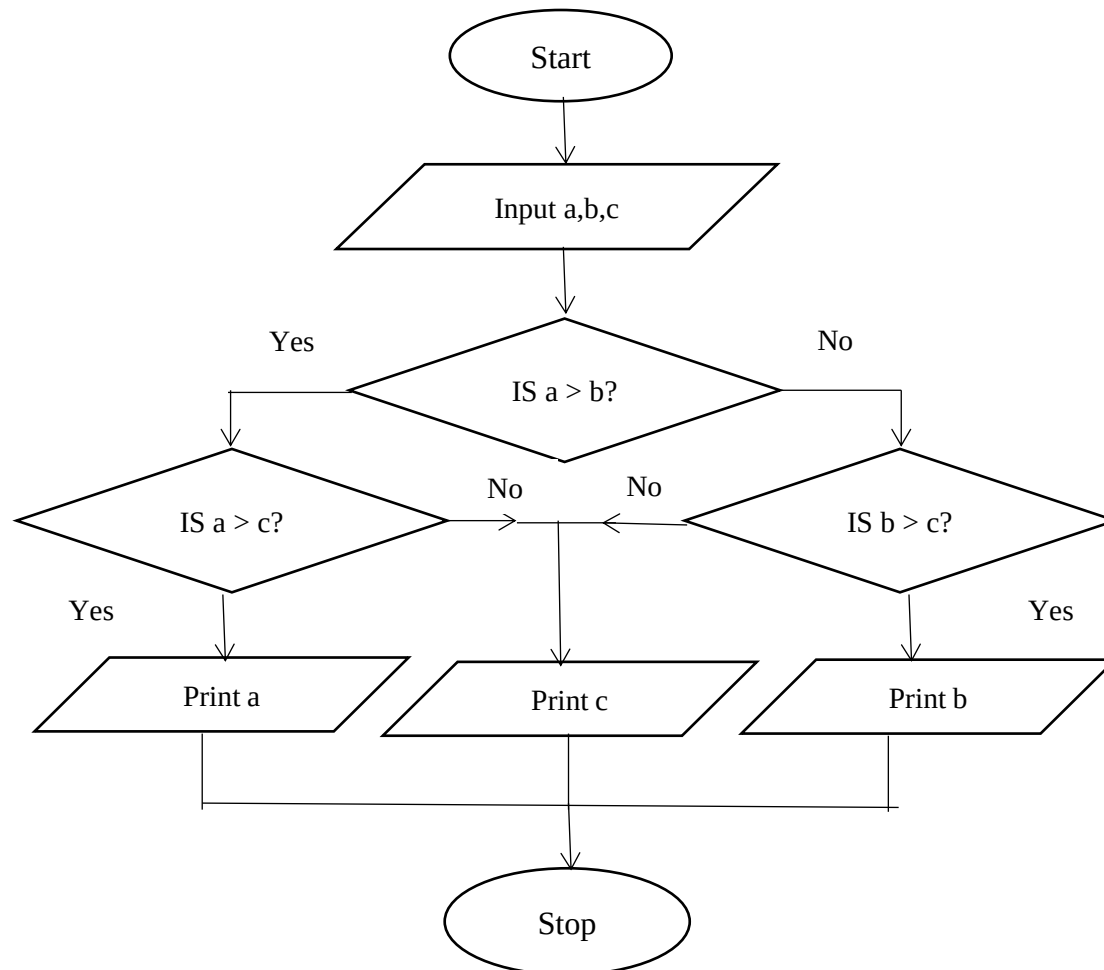
Draw the Flowchart to find the average of three numbers



- **Step 1:** Read the numbers a, b, c
- **Step 2:** Compute the sum of a, b and c
- **Step 3:** Divide the sum by 3
- **Step 4:** Store the result in variable d
- **Step 5:** Print the value of d
- **Step 6:** End

Example 2

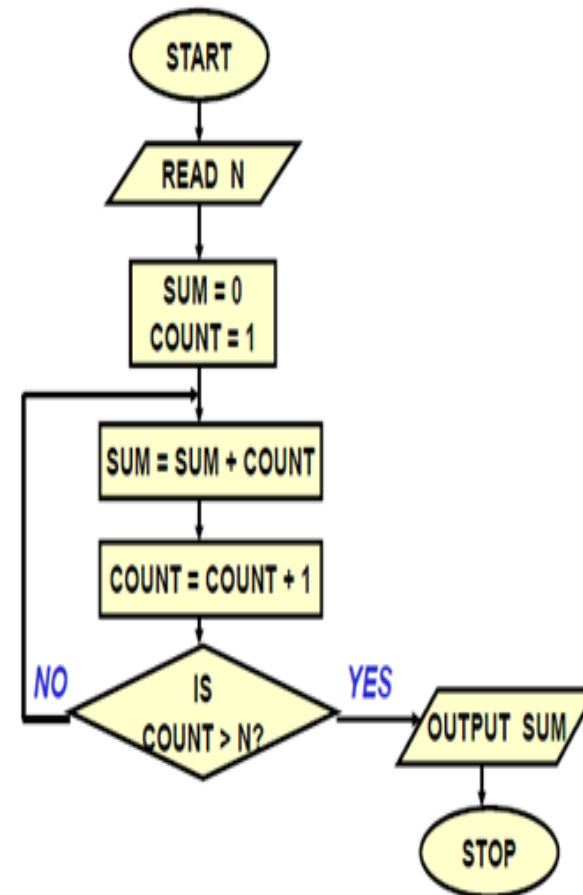
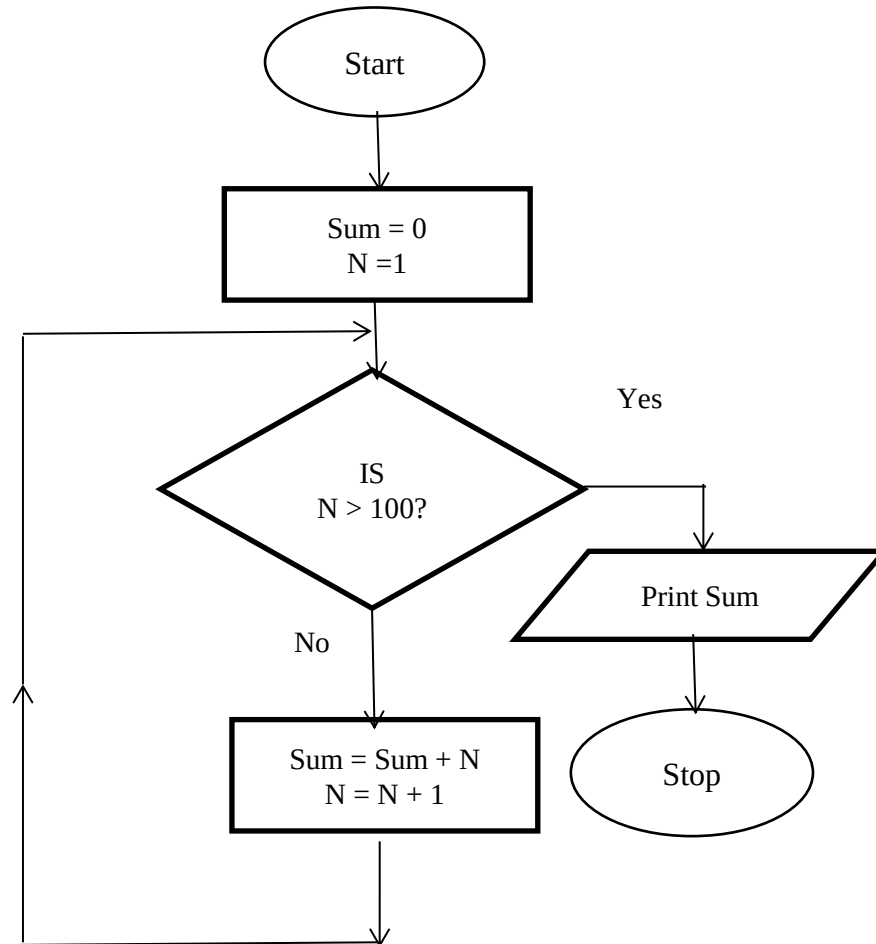
Draw the Flowchart to find out the biggest of the three unequal positive numbers.



- **Step 1:** Read the numbers X,Y,Z.
- **Step 2:** if ($X > Y$)
- $Big = X$
- else $Big = Y$
- **Step 3:** if ($Big < Z$)
- **Step 4:** $Big = Z$
- **Step 5:** Print the largest number i.e. Big
- **Step 6:** Stop.

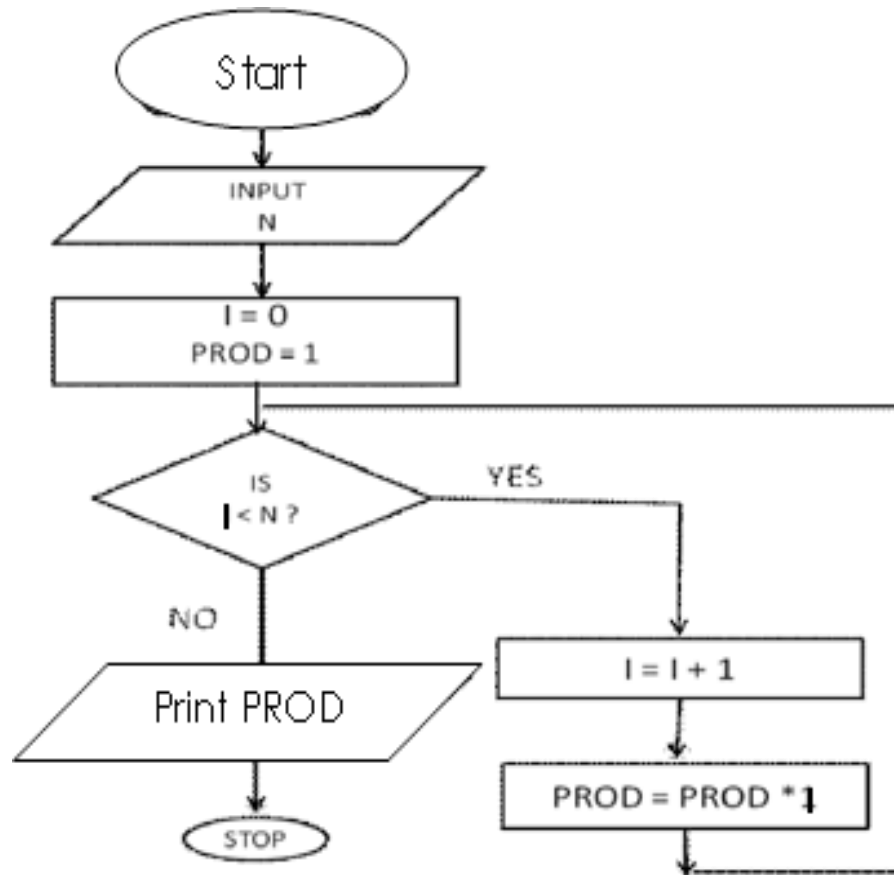
Example 3

Draw the Flowchart for adding the integers from 1 to 100 and to print the sum.



Example 4

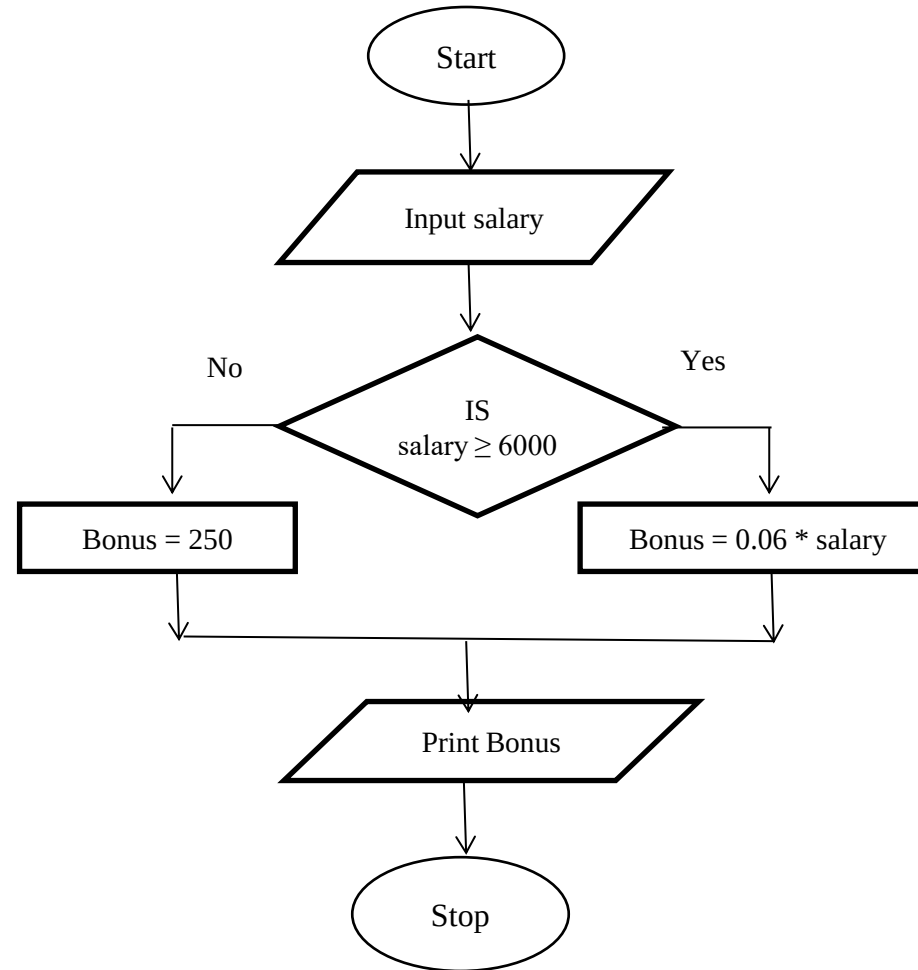
Draw a flowchart to find the factorial of given positive integer N.



- Step 1: read N
- Step 2: $I = 0$
- Step 3: $PROD = 1$
- Step 4: While $I < N$ do
 - $I = I + 1$
 - $PROD = PROD * I$
- Step 5: Write “Factorial of”, N, “is”, PROD
- Step 6: End

Example 5

ABC company plans to give a 6% year end bonus to each of its employees earning EGP 6,000 or more per month, and a fixed EGP 250 bonus to the remaining employees. Draw a flowchart for calculating the bonus for an employee



Example of Algorithms and Flowcharts

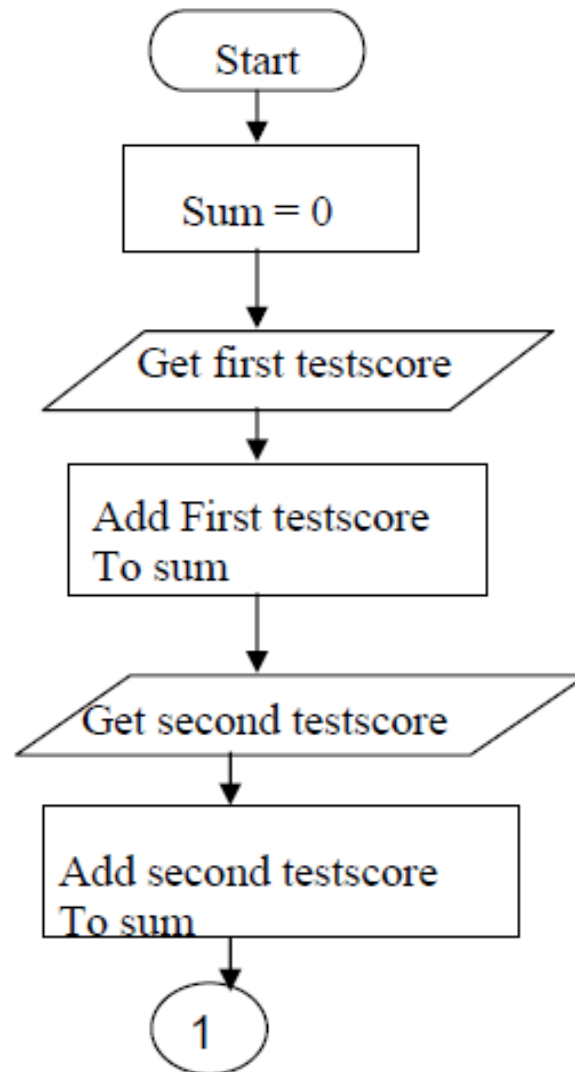
Design an algorithm and the corresponding flowchart for adding the test scores as given below:

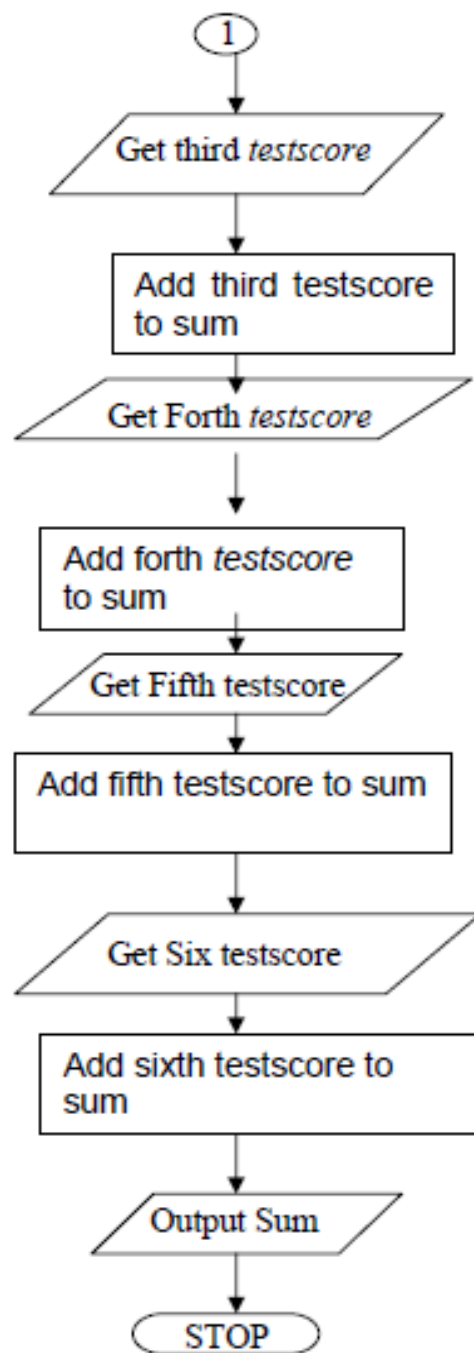
26, 49, 98, 87, 62, 75

a) Algorithm

1. Start
2. Sum = 0
3. Get the first testscore
4. Add first testscore to sum
5. Get the second testscore
6. Add to sum
7. Get the third testscore
8. Add to sum
9. Get the Forth testscore
10. Add to sum
11. Get the fifth testscore
12. Add to sum
13. Get the sixth testscore
14. Add to sum
15. Output the sum
16. Stop

B. Flowchart





The Pseudo code

- The Pseudo code is **neither an algorithm nor a program**. It is an abstract form of a program. It consists of English like statements which perform the specific operations. It is defined for an algorithm. It does not use any graphical representation. In pseudo code, the program is represented in terms of words and phrases, but the syntax of program is not strictly followed

Advantages of pseudo code

- Easy to read.
- Easy to understand.
- Easy to modify.

Example of pseudo code

Write a pseudo code to perform the basic arithmetic operations.

- Read $n1, n2$
- $\text{Sum} = n1 + n2$
- $\text{Diff} = n1 - n2$
- $\text{Mult} = n1 * n2$
- $\text{Quot} = n1/n2$
- Print sum, diff, mult, quot
- End.

Determining a Student's Final Grade

Algorithm

Write an algorithm and pseudocode to determine a student's final grade and indicate whether it is passing or failing. The final grade is calculated as the average of four marks.

Algorithm:

1. Ask the user to enter 4 marks (Mark1, Mark2, Mark3, Mark4)
2. Calculate the marks average (Avg) by summing marks and it dividing by 4
3. If average (Avg) is greater than or equal 60
4. Print "Pass"
5. Else
6. Print "Fail"
7. End if

Determining a Student's Final Grade

Pseudocode

Write an algorithm and pseudocode to determine a student's final grade and indicate whether it is passing or failing. The final grade is calculated as the average of four marks.

Pseudocode:

1. input Mark1, Mark2, Mark3, Mark4
2. $Avg = (Mark1 + Mark2 + Mark3 + Mark4) / 4$
3. if Avg $\geq$ 60:
 4. print "Pass"
5. else:
 6. print "Fail"
7. End if

Difference between Algorithm, Pseudocode and Program

- **Algorithm:** **Systematic logical approach**, which is a well-defined, step-by-step procedure that allows a computer to solve a problem.
- **Pseudocode :** It is a **simpler version of a programming code** in plain English which uses short phrases to write code for a program before it is implemented in a specific programming language.
- **Program :** It is **exact code** written for problem following all the rules of the programming language.